

[Physical World Modeling for Robot Manipulation]

Workshop

<https://phiwm.github.io/CORL2026.html>

Organizers

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Description

Robot manipulation is moving on from simple pick and place. New and more capable embodiments, end-effectors and sensors and advances in learning algorithms combined with large-scale data means we can now start tackling more difficult tasks: cutting a bowl of fruit with a knife, or cracking eggs and making an omelette. These tasks require delicate handling and complex tool use, and unlocking them requires models that predict and reason about physical states of agents and environments. World modeling is becoming a go-to paradigm for many aspects of research in robot manipulation: for planning, for policy evaluation and reinforcement learning, even as a backbone for generalist robot policies. Driven by the success of video-based and 3D-based world models, world models are increasingly built toward action-conditioned models for robot manipulation. However, visual observation alone is not enough for manipulation. Defining constraints for planning, formulating rewards for RL, and setting termination conditions for policy evaluation are just some examples of aspects of robot manipulation research that require access to physical states of the agent and the environment that current world models do not provide. Cutting fruits or cracking eggs demands knowing how much force to apply and when to back off. Apply too much and you damage it, too little and you lose control. To cook properly, it need to actively sense the temperature. This workshop asks: what physical states must a world model represent to support real manipulation? Thus, we propose this workshop **PhysicalWM: Physical world modeling for robotic manipulation** as a key next step toward embodied intelligence.

A **physical world model** should go beyond visual appearance and geometric reconstruction. It should represent physical-aware states: how objects can move and interact under contact and manipulation, which actions are feasible, and which future states may result. This requires reasoning about force, articulation, material properties, and so on from sensor data including and beyond RGB and depth cameras. Despite rapid progress, physical world modeling remains at an early stage. Researchers working on topics like video generation, 4D scene representations, and generalist robot policies, are approaching the problem from different perspectives. However, these communities often use different assumptions, representations, datasets, and evaluation protocols. A dedicated workshop is therefore needed to clarify how these threads should be connected.

Related Workshops and Differentiation

Recent workshops have explored complementary aspects of world modeling for embodied agents. *Structured World Models for Robotic Manipulation* at RSS 2025 studied structured representations, simulation, planning, and policy learning [4]. The CoRL 2025 workshops *Learning to Simulate Robot Worlds* and *Robotics World Modeling* focused on learned simulation and robot-centric world models [2, 3]. *Embodied World Models for Decision Making* at NeurIPS 2025 considered broader topics including model-based reinforcement learning, long-horizon planning, interactive generation, and VLA models [1]. The CVPR 2026 workshop *World Models Meet Active Sensing and Closed-Loop Planning* connected world models with active perception and closed-loop embodied tasks [5].

Our PhysicalWM workshop is distinguished by its focused question: **what action-relevant physical state beyond visual inputs should a world model represent for robot manipulation?** The workshop emphasizes multimodal physical signals, including tactile, forces, thermal, and proprioceptive observations and evaluation through downstream planning and control for robot manipulation. Its intended outcome is a community-authored physical world model state and evaluation protocol for physically grounded robot manipulation.

Core Challenges / Research Questions

The workshop focuses on one central question: *What physical state should an action-conditioned world model represent to improve decision-making in robot manipulation, and how should this capability be evaluated?*

We organize the workshop around three connected research questions:

1. **Physical-state representation.** Which decision-relevant quantities beyond RGB-D, such as contact, force, compliance, articulation, and uncertainty, should be represented explicitly or implicitly in a physical world model?
2. **State inference and prediction.** How can these physical states and their action-conditioned transitions be learned from partially observed multimodal interaction data when direct supervision is limited?
3. **Manipulation-oriented evaluation.** How should physical world models be evaluated based on prediction accuracy, physical consistency, uncertainty, and their downstream utility for planning and control in contact-rich manipulation?

Invited Speakers

All confirmed speakers have been informed of CoRL’s two-workshop participation limit and will be encouraged to attend the full workshop and co-facilitate the breakout session.

Ming-Yu Liu

NVIDIA

Omnimodal World Models for Physical AI

COMMITTED

mingyul@nvidia.com

Manling Li

Northwestern University

Multimodal embodied agents: building systems that connect language, vision, and action to support grounded reasoning, physical-world understanding, and interactive planning.

COMMITTED

manling.li@northwestern.edu

Tamim Asfour

KIT

COMMITTED

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Developing humanoid systems that combine perception, manipulation, learning, and human-robot interaction to operate in complex real-world environments.

Saining Xie

AMI Labs & NYU

REQUESTED

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World models for embodied intelligence: toward systems grounded in continuous sensory experience and real-world interaction.

Proposed Format and Interactivity

PhysicalWM will be a half-day, in-person workshop combining interactive keynotes, lightning pitches, posters, live demonstrations, and structured community discussion. We will solicit non-archival short papers, including position papers, work in progress, negative results, and benchmark proposals, as well as demo submissions featuring physical robot systems, multimodal sensing platforms, executable world-model prototypes, and interactive visualizations.

The program will begin with brief opening remarks followed by four 30-minute interactive keynotes. Each keynote will consist of a 25-minute presentation and a 5-minute moderated discussion centered on a concrete open question. Invited speakers will be encouraged to attend the full workshop and help facilitate the subsequent discussions.

Selected contributors will give brief lightning pitches before an integrated poster and live-demo session. This session will allow participants to interact directly with authors, inspect working systems, and compare approaches to physical-state representation, action-conditioned prediction, and manipulation-oriented evaluation.

Following the coffee break, participants will join structured breakout groups to co-design benchmarks and evaluation criteria for physical world models. Each group will use a common worksheet and report its conclusions during a community synthesis session. The resulting recommendations will contribute to the community-authored *Physical World Model State and Evaluation Protocol*. The workshop will conclude with the Best Paper and Best Demo awards and closing remarks.

Contributed Papers Plan

We will solicit **non-archival workshop submissions** through the workshop website, mailing lists, relevant research communities, and social media. Submissions may include short papers, position papers, work-in-progress papers, negative results, benchmark or dataset proposals, system descriptions, and papers showing impressive recent progress or strong preliminary results.

The workshop will include a **poster track** and a **demo track**. The poster track will focus on physical state representation, multimodal sensing, action-conditioned prediction, robot manipulation, simulation-to-real transfer, evaluation protocols, and embodied datasets. The demo track will invite interactive systems such as robot demonstrations, executable world-model prototypes, multimodal sensing platforms, manipulation benchmarks, and visualization tools.

All submissions will be reviewed by the organizers and a program committee with expertise in computer vision, machine learning, and robotics. Reviews will consider novelty, potential impact, and technical quality. Three submissions from each track will be selected for short oral presentations. The workshop will also recognize one **Best Paper** and one **Best Demo**.

Plans to Encourage Participation and Diversity

We will encourage participation beyond the invited speaker list through an open call for **non-archival submissions**, including short papers, position papers, work-in-progress results, negative results, benchmark proposals, and demo submissions. The poster and demo tracks are designed to broaden participation beyond traditional paper presentations, giving students, early-career researchers, system builders, and interdisciplinary contributors multiple ways to engage

with the workshop. Selected submissions from both tracks will receive oral presentations, and the poster/demo session and panel discussion will provide additional opportunities for direct interaction with speakers, organizers, and attendees. We will recognize outstanding contributions through **Best Paper** and **Best Demo** awards, with prizes subject to available sponsorship. To support real-world demonstrations, we will provide travel support for selected demo presenters, particularly when transporting embodiments or hardware would otherwise create a barrier to participation.

Our organizing team is itself diverse across career stages, sectors, institutions, and geographies. It includes PhD students, a postdoctoral researcher, assistant professors, full professors, and an industry research scientist, with broad international representation across seven countries and three continents: Asia, Europe, and North America. We will use this network to disseminate the call broadly across robotics, computer vision, machine learning, simulation, and embodied AI communities. We will also actively promote diversity in the workshop program. We will encourage submissions from junior researchers and underrepresented communities, and ensure that the panel and discussion format gives meaningful opportunities to contribute to other participants than the invited speakers.

Recommended Program

The half-day workshop combines invited talks, interactive demonstrations, and structured community discussion. Each 30-minute keynote consists of a 25-minute presentation followed by a 5-minute moderated discussion.

Table 1: Proposed half-day workshop schedule.

Time	Activity
00:00–00:10	Opening and challenge framing
00:10–01:10	Two keynote speakers (25-minute talk + 5-minute discussion)
01:10–01:45	Poster session, live demonstrations and coffee break
01:45–02:00	3 oral presentation sessions (each for 5 min)
02:00–03:00	Two keynote speakers (25-minute talk + 5-minute discussion)
03:00–03:35	Structured breakout: benchmark and evaluation-protocol design
03:35–03:50	Group reports, discussion, and community synthesis
03:50–04:00	Best Paper and Best Demo awards; closing remarks

Sponsorship

Our workshop will be sponsored by Litchi AI, a company focused on world modeling and Physical AI infrastructure. The sponsorship will support the main workshop costs, including awards, demo logistics, and limited travel support for real-world demo presenters. All review, selection, award, and program decisions will remain independent and under the responsibility of the organizers.

Logistics and Resources

Estimated audience size: We expect approximately 80–120 in-person participants. If virtual participation or livestreaming is supported by the conference, we expect an additional 50–100 remote attendees.

Physical requirements: We request a standard workshop room with projector, screen, microphones, speakers, and reliable Wi-Fi. For the poster and demo session, we request poster boards for accepted poster-track papers and several tables for demo-track presentations. Demo tables should

ideally have nearby power outlets or power strips. If space allows, a small open area would be useful for lightweight robot or interactive system demonstrations.

Special requirements: No major special requirements are needed beyond standard Audio-Visual support, poster boards, demo tables, power access, and Wi-Fi. For safety and logistics, any physical robot demos will be limited to compact, tabletop, or otherwise conference-safe setups.

Commitments

- ✓ At least one organizer will attend CoRL 2026 in person.
- ✓ Organizers will attend the CoRL 2026 workshop organizers meeting.
- ✓ Organizers will submit a post-workshop artifact.
- ✓ Willingness to merge with related proposals if requested.

References

- [1] Embodied world models for decision making. NeurIPS Workshop, 2025.
- [2] Learning to simulate robot worlds. CoRL Workshop, 2025.
- [3] Robotics world modeling. CoRL Workshop, 2025.
- [4] Structured world models for robotic manipulation. RSS Workshop, 2025.
- [5] World models meet active sensing and closed-loop planning. CVPR Workshop, 2026.